The PTC[ASYNC] bit is an optimization of the transfer and is subject to negotiation with the Responder. In general if the Initiator proposes an ASYNC transfer, it SHOULD also be prepared to accept a synchronous transfer, and SHOULD at least list one of PTC[RECEIVER_DRIVE] or PTC[SENDER_DRIVE] in the request.

Multiple transfer modes can be specified, which signifies that the Initiator can support any of those transfer modes. For example, if PTC[ASYNC], PTC[RECEIVER_DRIVE] and PTC[SENDER_DRIVE] bits are set on a SendInit, it indicates that the Initiator supports 3 distinct transfer modes: synchronous Sender drive, synchronous Receiver drive and asynchronous (drive is implied). Only one transfer mode SHALL be used in the actual transfer. The Responder SHALL choose which one to use. If the Responder supports both Sender and Receiver drive, it SHOULD prefer to use Sender drive to retain the request/response semantics.

NOTE Support for the asynchronous mode is provisional and SHALL not be chosen by the Responder.

Table 111. SendInit/ReceiveInit Proposed Transfer Control (PTC) field structure

bit 7	6	5	4	3	2	1	0
RFU	ASYNC	RECEIVER_DRIVE	SENDER_DRIVE	VERSION			

The fields for PTC are:

SendInit/ReceiveInit PTC[VERSION]: proposed protocol version

- Width: 4 bits
- Values: 0x00-0x0F proposed protocol version

SendInit/ReceiveInit PTC[SENDER_DRIVE]: sender drive supported

- Width: 1 bit
- Values:
  - 0 if Sender drive is not supported
  - 1 if Sender drive is supported by Initiator

SendInit/ReceiveInit PTC[RECEIVER_DRIVE]: receiver drive supported

- Width: 1 bit
- Values:
  - 0 if Receiver drive is not supported
  - 1 if Receiver drive is supported by Initiator

SendInit/ReceiveInit PTC[ASYNC]: asynchronous mode supported

- Width: 1 bit

